NETWORKING FUNDAMENTALS

IMAP (Internet Message Access Protocol)

Computer Networking

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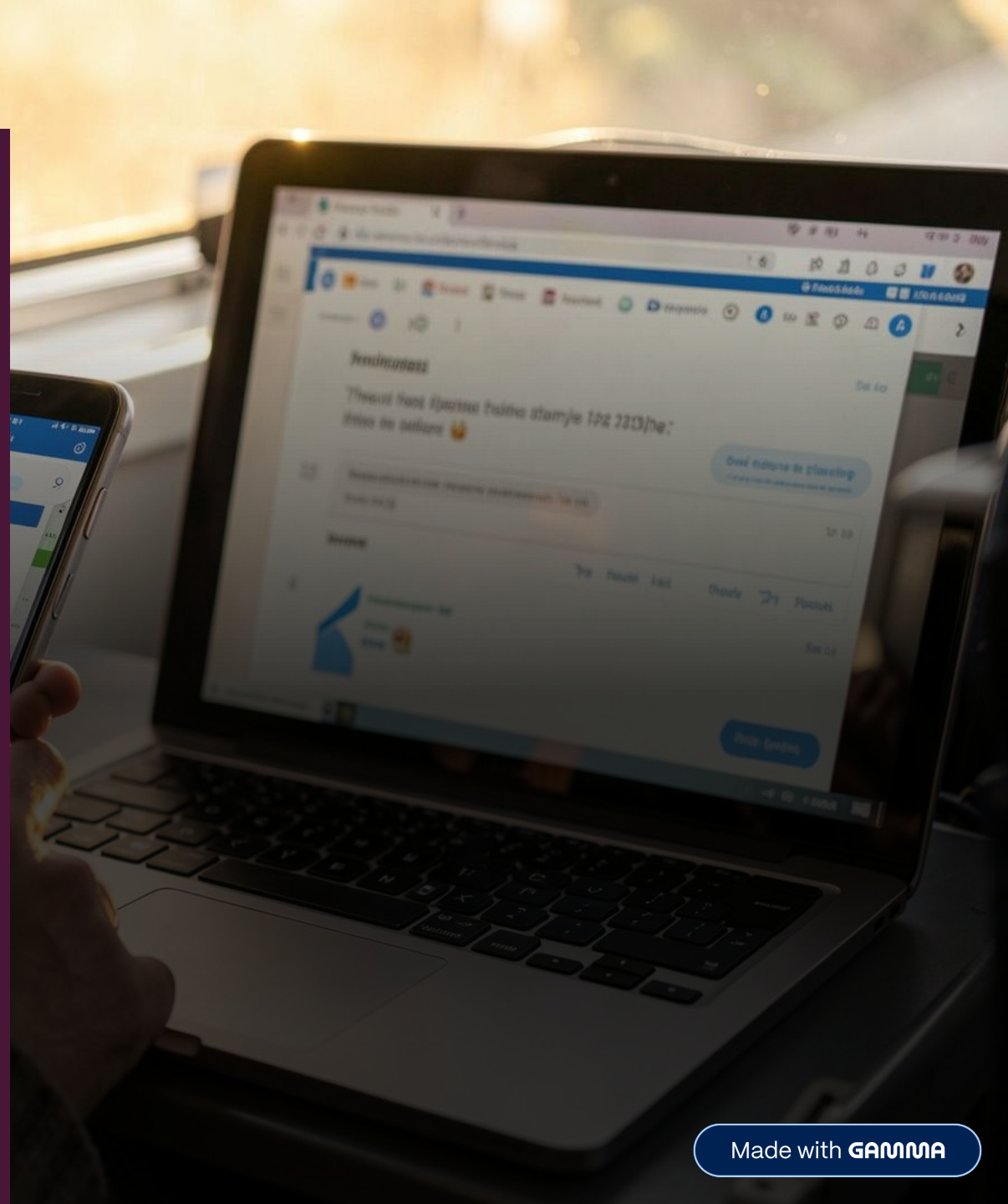
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IMAP: how your email stays in sync across all devices

A simple lesson on server-side mailbox access, sync and secure connections

A 10-minute classroom presentation



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Where does IMAP fit in the email system?

SMTP

The protocol used to send mail.

IMAP

Reads, organises and syncs mail stored on the server.

Mailbox

The mail client talks to the mail server that holds the mailbox.



CLIENT · SERVER · MAILBOX

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What is IMAP?

IMAP (Internet Message Access Protocol) is how a mail client reads and manages mail stored on a mail server.

Access

Read, search and organise mail that stays on the server.

Sync

Keep every device aligned with the same mailbox state.

Port

Usually 143 with STARTTLS, or 993 with TLS.

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The core idea of IMAP: mail lives on the server

With IMAP, a phone or laptop is not the only owner of the mailbox. The master mailbox stays on the server; devices view the same state and keep local copies when needed.



Folders

Inbox, Sent, Drafts and custom folders all stay on the server.



One shared mailbox

Phone, laptop and webmail can all see the same mailbox state.



State sync

Read, unread, flag and delete status sync on the server.

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SMTP's role: sending mail

SMTP moves outgoing mail from your client to a server, and between mail servers. It does not keep your inbox.



Compose

You write and send from the mail client.



Submit

SMTP hands the message to the outgoing server.



Relay

Servers pass the message toward the recipient's server.



Deliver

The recipient's server stores it, ready for IMAP or POP3.

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POP3: download, then usually remove

POP3 is the older retrieval model: the client downloads messages to one device, and the server copy is often deleted.

Download

Messages are copied to the client device.

Local archive

The device becomes the main place where mail is kept.

Limited sync

Folders and read status rarely match across devices.

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How does an IMAP session work?

The client first opens a connection to the server, then selects a mailbox and syncs the data it needs.



Connection

The client sets up a secure connection.



Authentication

Access is granted once identity is verified.



Select

The client selects a mailbox, such as INBOX.



Fetch & sync

Fetches headers or bodies; syncs folders and flags.

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The IMAP workflow in commands

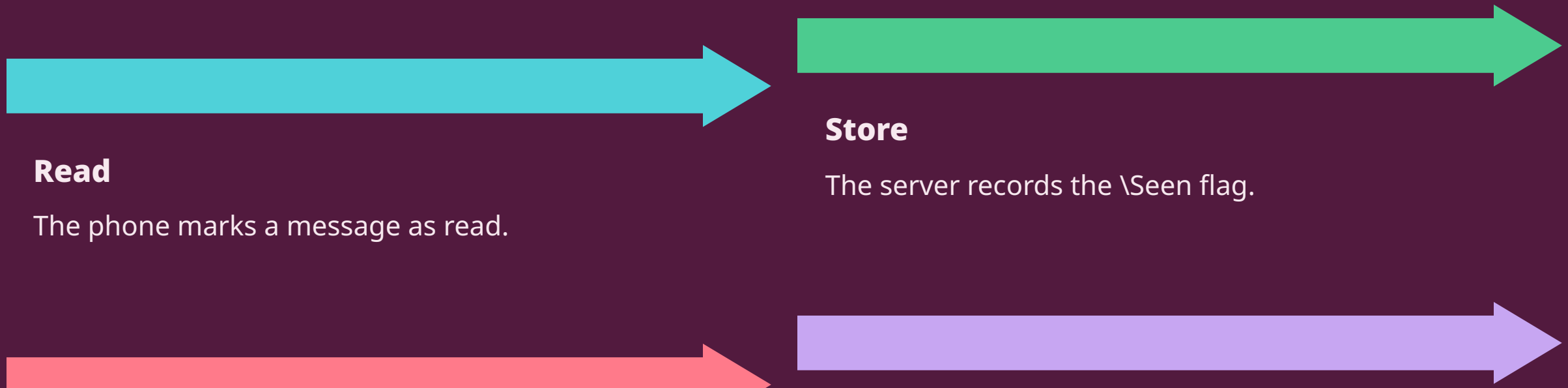
A client drives the session with a few core commands. Each one changes what the client can see.

Command	Purpose
LOGIN / AUTHENTICATE	Prove identity before access
SELECT INBOX	Open a mailbox for work
FETCH	Download headers or message bodies
STORE	Change flags, such as \Seen
SEARCH	Find messages on the server
LOGOUT	End the session

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Mailbox synchronization

One action on one device becomes visible everywhere, because the server holds the shared state.



Read

The phone marks a message as read.

Store

The server records the \Seen flag.

Fetch

The laptop checks the mailbox and sees the change.

Match

Webmail shows the same message as read.

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Folders and flags live on the server

IMAP keeps the folder tree and message markers in one place, so every client sees the same organisation.

Folders

Inbox, Sent, Drafts, Trash and custom folders, including nested ones.

\Seen and \Flagged

Read status and starred importance travel with the message.

\Answered and \Draft

Reply state and unsent drafts are shared across devices.

\Deleted and EXPUNGE

\Deleted marks mail; EXPUNGE removes it from the mailbox.

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IMAP vs POP3: the most important difference

With one account on several devices, the key question is where the mailbox's authoritative state should live.

Topic	IMAP	POP3
Where mail lives	Mainly on the server	Usually downloaded to the client
Multiple devices	Suited to syncing	Built around one main device
Folders and flags	Managed on the server	Limited
Offline use	Possible for previously synced items	Downloaded mail stays local

SHORT RULE · IMAP is sync-based, POP3 is download-based

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One message, three devices

Ana reads a message on her phone in the morning. Here is what each protocol does next.

Step	IMAP	POP3
Read on phone	Server marks the message as read	Message is downloaded and removed from the server
Open laptop	Same message, already read	Laptop sees nothing new
Reply from laptop	Sent copy syncs everywhere	Sent copy stays on the laptop only

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Ports and security: 143 vs 993

A port is only the door to a connection; security depends on how TLS and proper authentication are used at that door.

STANDARD IMAP

143

STARTTLS

The standard IMAP port. STARTTLS can upgrade the connection to encrypted.

IMPLICIT TLS

993

SSL/TLS

The connection is encrypted from the start; the common port for secure IMAP.

Authentication

Where the provider supports it, OAuth 2.0 can be a safer choice than a password.

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Two ways to encrypt IMAP

Both reach a secure result, but they start encryption at different moments.

PORT 143

STARTTLS

The client connects in plain text, then asks the server to upgrade to TLS. If the upgrade is skipped, the session can stay unencrypted.

PORT 993

Implicit TLS

The connection is encrypted from the first byte, so there is no unencrypted window.

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TLS protects the connection

Transport Layer Security (TLS) encrypts traffic between the client and the server, so passwords and messages cannot be read in transit.

Confidentiality

Eavesdroppers see scrambled data, not message content.

Integrity

Tampered data is detected and rejected.

Server identity

The certificate helps confirm the client is talking to the real mail server.

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Authentication: proving who you are

Before any mailbox is opened, the server checks the user's identity.

Password login

Simple, but risky if reused or sent without TLS.

App passwords

Separate passwords for mail apps, each revocable on its own.

OAuth 2.0

Token-based sign-in, often safer where the provider supports it.

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A secure IMAP setup checklist

Use this list whenever you configure an IMAP account.

- Use port 993 with SSL/TLS, or port 143 with STARTTLS
- Refuse unencrypted connections and do not ignore certificate warnings
- Use OAuth 2.0 or an app password where available
- Turn off legacy plain-text authentication
- Review connected devices and revoke any you no longer use

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Setting up IMAP on a phone

The same four settings work across most mail apps.



Account

Choose IMAP as the account type.

Server

Enter the incoming server and port 993.



Security

Select SSL/TLS, then sign in.

Folders

Choose which folders to keep in sync.

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Typical incoming settings

Exact host names vary by provider; the pattern stays the same.

Setting	Typical value
Protocol	IMAP
Port (secure)	993
Encryption	SSL/TLS
Authentication	OAuth 2.0 or password
Server name	imap.example.com (example only)

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Advantages of IMAP

Why many users and organisations choose it.

Multi-device sync

One mailbox state on every device.

Server-side folders

Organisation is shared and consistent everywhere.

Partial fetch

Headers can load first; full bodies load on demand.

Server search

Search the whole mailbox from any client.

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Limitations of IMAP

IMAP is powerful, but it depends on the server and the network.

Server dependence

Without the server, the mailbox is unavailable.

Storage use

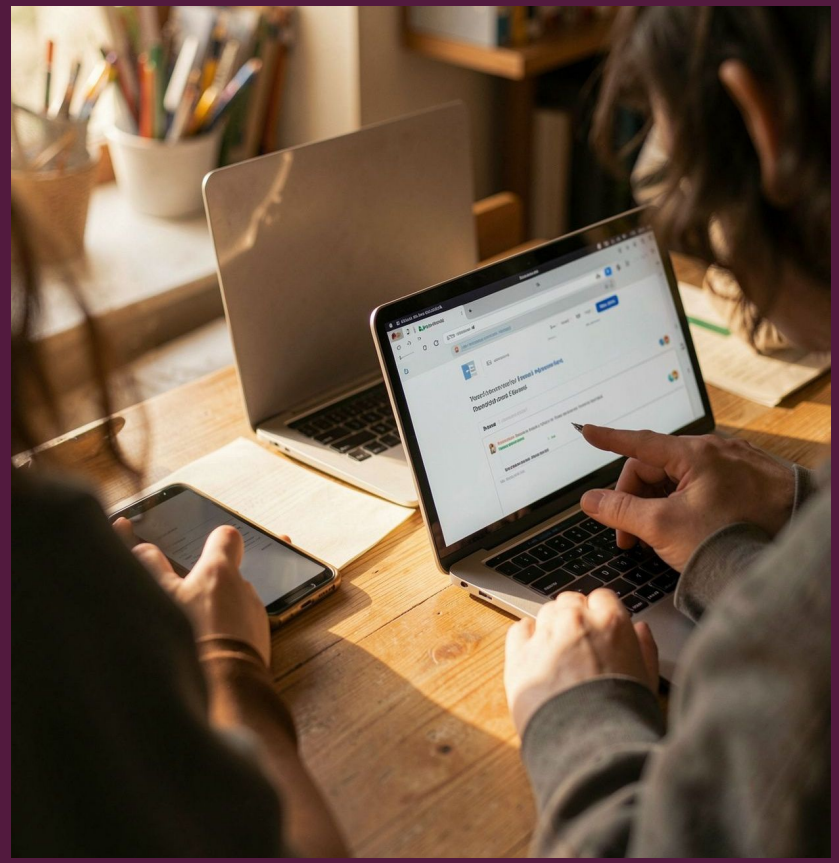
Full mailboxes and local caches use server or device space.

Offline gaps

Only previously synced messages are available offline.

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When should you choose IMAP?



ONE ACCOUNT · MANY DEVICES

Choose IMAP when you want the account's mailbox to stay consistent across several places.

One account, many devices

Use the same mailbox from your phone, laptop and webmail.

Keep folders and read status the same

Sent mail, folders, and read and flag states should match everywhere.

Centralized search needed

You want flexible search and central access to the mailbox on the server.

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Common mistakes and fixes

Most IMAP problems come from a few predictable settings.

Mistake	Fix
Port 143 with no STARTTLS	Switch to port 993 with SSL/TLS
Expecting POP3-style downloads	Remember that IMAP keeps mail on the server
Deleting only the local copy	Delete on the server so the change syncs everywhere
Ignoring certificate warnings	Check the server name before accepting

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IMAP vs POP3 in one view

Keep these three contrasts in mind.

Storage

IMAP: on the server. POP3: usually on the device.

Sync

IMAP: many devices share one state. POP3: one main device.

Security

Both need TLS; IMAP usually uses port 993, POP3 uses 995.

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TAKEAWAY

Remember IMAP: mailbox on the server, state in sync.

IMAP means server-side mailbox access and multi-device synchronisation; POP3 means download-based retrieval. Configuring ports, TLS and authentication correctly is the basis of safe use.

Server

State of the master mailbox

Sync

The same view on every device

Secure

Port 993, TLS and proper auth

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